bAI buddies Project Documentation

Project overview

bAI buddies is a web application that provides conversational fitness and nutrition guidance. It has two coach modes: **FitBuddy** for workout support and **MealBuddy** for meal and nutrition support. Users can type messages or use voice controls to ask questions, get suggestions, and view generated plans.

Purpose

The project aims to make basic workout and meal planning easier to access through a guided, friendly interface. FitBuddy considers the user's fitness goals, experience, preferences, and available exercise equipment when suggesting workouts. MealBuddy helps users explore food preferences, recipes, and meal plans. The application is an educational aid and does not replace professional fitness, nutrition, or medical advice.

Main features

- Separate fitness and nutrition coach experiences.
- Text chat and microphone-based voice interaction, with spoken coach responses.
- Workout suggestions and plans filtered against available equipment and user preferences.
- Nutrition conversations and meal-plan responses.
- User, equipment-profile, and workout-plan storage in a database.
- AI-generated responses when a provider is configured, with a mock fallback for demonstrations and local use.

How the application works

1. A user opens the web interface and chooses a coach mode.
2. They ask a question by typing or speaking and provide relevant goals or preferences.
3. The frontend sends the conversation to the Python API. For workout plans, the backend checks the available exercise list against the user's equipment and preferences.
4. The configured AI provider, or the mock fallback, prepares a response or plan.
5. The app displays the result and can play the response as speech. User profiles, equipment profiles, and workout plans are stored by the backend; chat and plan display state is also kept in browser storage.

Project structure

- `frontend/` — HTML, CSS, and JavaScript for the landing page, coach interface, chat, and voice controls.
- `backend/` — FastAPI endpoints, AI response generation, database models, and data handling.
- `tests/` — automated checks for application logic.
- `fitbuddy.db` — local SQLite database file used by default.
- `requirements.txt` — Python package dependencies.

Technology summary

The frontend uses plain HTML, CSS, and JavaScript. The backend uses Python, FastAPI, and SQLAlchemy. SQLite is the default database. AI responses can use configured Groq or Gemini credentials; if live AI is unavailable or mock mode is selected, the application uses its built-in mock behavior. Voice input and playback are provided through the browser and the backend speech endpoint.

Running locally

Install the packages listed in `requirements.txt`, configure environment variables in `.env` if using a live AI provider, then start the API from the project directory:

```
uvicorn backend.main:app --reload
```

Open `http://localhost:8000/` in a browser. Interactive API documentation is available at `http://localhost:8000/docs` while the server is running.